

Qn	Ans	Solution
5	B	Let the tension in string be T and assume the 500 g block accelerates downwards with a. Consider the free body of each block, $0.500(9.81) - T = 0.500a \dots\dots\dots (1)$ $T - 0.300(9.81) = 0.300a \dots\dots\dots (2)$ (1) + (2): $0.5(9.81) - 0.3(9.81) = 0.8a$ $a = 2.5 \text{ m s}^{-2}$
6	B	By Principle of Conservation of momentum.